

INTERIM RESULTS: PER-ROLLOUT CALIBRATION OF THE FREE EMA-KL SURROGATE UNDER HETEROGENEOUS STALENESS

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ABSTRACT. Asynchronous RL pipelines mix rollouts generated by policy snapshots of different ages Δ_i , but the free EMA-KL surrogate $c(\log T_n - \log T_{n-\Delta})$ with $c = \alpha/(\Delta(1-\alpha))$ has so far only been validated at a single, homogeneous Δ . We add a heterogeneous-staleness mode to tba-prime (per-DP-rank checkpoint lags) and a per-rollout coefficient $c_i = \alpha/(\Delta_i(1-\alpha))$, so that every rollout estimates the same $\beta \text{KL}(T_n \| E_n)$ regardless of its age. Interim findings: (i) a homogeneous $\Delta=10$ reference run through the new mechanism reproduces the historical no-centering baseline (peak eval 0.829 vs. 0.827–0.831); (ii) the calibrated heterogeneous arm, $\Delta_i \in \{1, 4, 10, 32\}$, tracks that reference essentially point-for-point (eval 0.816 vs. 0.814 at matched step 520) despite a $32\times$ staleness spread; (iii) per- Δ telemetry shows the surrogate’s relative error is strongly Δ -dependent (median rel. error of means 5.9 at $\Delta \leq 2$ vs. 0.19 at $\Delta \geq 21$), empirically confirming the finite-difference variance warning in the theory note — yet training is robust to the noisy small- Δ estimates. Symmetric NVFP4-emulated (QDQ) training is stable through a 60-step smoke. Dose-sweep and $\beta=0$ controls, which decide whether *miscalibration* hurts, are queued.

1. SETUP

All runs train Qwen3-4B-Instruct-2507 on Countdown with TBA (no-centering, importance sampling on), lr 10^{-6} , $K=16$ rollouts/problem, sequence length 4096, 720 target steps, evaluation every 20 steps, EMA decay $\alpha = 0.9$, on 2 nodes ($4\times\text{H100}$ inference, $4\times\text{H100}$ training). Full config listings are in Appendix A.

Heterogeneous staleness. The inference side runs 4 data-parallel vLLM workers; `--rl.staleness-offsets` $o_{0..3}$ makes rank r reload, every step, the checkpoint from o_r steps ago. Each problem’s K samples come from one rank (policy-homogeneous groups); the training batch pools ranks, so batches mix policy ages. Verified in the rollout parquets: the trainer’s step-10 batch mixes generating checkpoints $\{9, 6, 0, 0\}$, i.e. the designed lags with the offset-32 rank still in its burn-in clamp.

Per-rollout calibration. Each rollout row records its generating checkpoint step; the trainer computes Δ_i per sample and, with `--grpo.kl-approx-delta-source per_rollout`, uses $c_i = \alpha/(\Delta_i(1-\alpha))$ (clamped $\Delta_i \geq 1$) in the approx-KL penalty. To first order every rollout then contributes an estimate of the *same* $\beta \text{KL}(T_n \| E_n)$; the `global` mode (one c from `max_async_level`) is the miscalibrated comparison arm. Both the importance-sampling anchor and the KL anchor are the sampler’s own vLLM logprobs (`--use-vllm-logprobs`) in every arm, per the design review: recomputing anchors from a single lagged snapshot is incoherent when ages are mixed. The per-token clamp is applied in raw log-ratio units (± 10 *before* the coefficient) so truncation is Δ -invariant.

2. RESULTS

Finding 1: the reference reproduces the historical baseline. The homogeneous $\Delta=10$ arm peaks at 0.829 (final 0.828 at step 680). The historical no-centering EMA cells B/D from the ema.KL phase peaked at 0.827–0.831 under the recompute anchor and the legacy reload rule. The new mechanism (per-step reloads, vLLM anchor, raw-units clamp) therefore changes nothing about the known-good baseline — the control the design review required before any cross-arm comparison.

Date: July 15, 2026.



FIGURE 1. Countdown eval accuracy. Blue: homogeneous $\Delta=10$ reference (run through the same offsets mechanism and anchor as the heterogeneous arms); aqua: heterogeneous $\Delta_i \in \{1, 4, 10, 32\}$ with per-rollout c_i ; yellow: heterogeneous with the KL penalty off ($\beta=0$; partial run, resubmitted after an unrelated time-limit stop). Runs end where their 24 h allocations ended (steps 688 and 561).

Finding 2: calibrated heterogeneity matches homogeneity. The decisive arm trains on batches whose rollout ages span $32\times$, yet its eval curve tracks the reference nearly point-for-point (at matched step 520: 0.816 vs. 0.814; peak 0.816). This is the “calibrated $\approx$ homogeneous” half of the calibration claim. The other half — that a *fixed* coefficient degrades under the same mixture — awaits the queued uniform-dose sweep ($w \in \{0.0045, 0.015, 0.045\}$) and the completed $\beta=0$ control. Notably, the partial $\beta=0$ curve is at *parity* with the calibrated arm through step 220 (e.g. 0.766 vs. 0.764 at step 200): heterogeneous staleness alone shows no early damage, consistent with the historical pattern that KL-related instabilities on this task surface after step ~ 300 . Whether the KL penalty earns its keep here is therefore still an open question that the completed controls will answer.

Finding 3: surrogate quality is strongly Δ -dependent — and training doesn’t care. The per- Δ telemetry (Fig. 2) shows median relative error of means of 5.88 ($\Delta \leq 2$), 2.13 ($\Delta \in [3, 6]$), 0.48 ($\Delta \in [7, 20]$), and 0.19 ($\Delta \geq 21$). This is exactly the bias-variance picture in the theory note’s limitations section: at small Δ the finite-difference slope is dominated by step-to-step SGD noise, and multiplying it by $c_i \propto 1/\Delta_i$ amplifies it (per-token MAE 0.018 at $\Delta \leq 2$ vs. 0.003 at $\Delta \geq 21$). Two implications. First, the $1/\Delta$ *mean* conversion is not refuted — the bias-oriented metrics shrink with Δ as the theory predicts — but small- Δ rollouts contribute high-variance KL estimates. Second, despite a quarter of tokens carrying rel. error ~ 6 estimates, the calibrated arm trained indistinguishably from the reference: the penalty’s group-summed, β -scaled form appears robust to zero-mean estimator noise. A practical refinement suggested by this data is a variance-aware clamp on c_i (e.g. flooring Δ_i at ~ 4), which we have not yet tested.

Finding 4: NVFP4-emulated symmetric training is stable (smoke scale). With QDQ (quantize-dequantize to the NVFP4 grid: E2M1 values, block-16 E4M3 scales) applied bit-exactly to both the trainer’s linear-layer forwards (STE) and the sampler’s checkpoints, 60-step smokes on one and two nodes ran without instability; eval reached 0.62–0.63 by step 40, consistent with BF16 runs at that depth. Saved checkpoints verified on-grid (99.97% of elements exactly; the residual 0.03% is a 1-ulp GPU/CPU float8 tie-rounding artifact of the checker, not of the contract, since both QDQ applications run on GPU). The FP4 collapse/rescue arms (staleness $\times$ quantization) are queued behind the calibration arms.

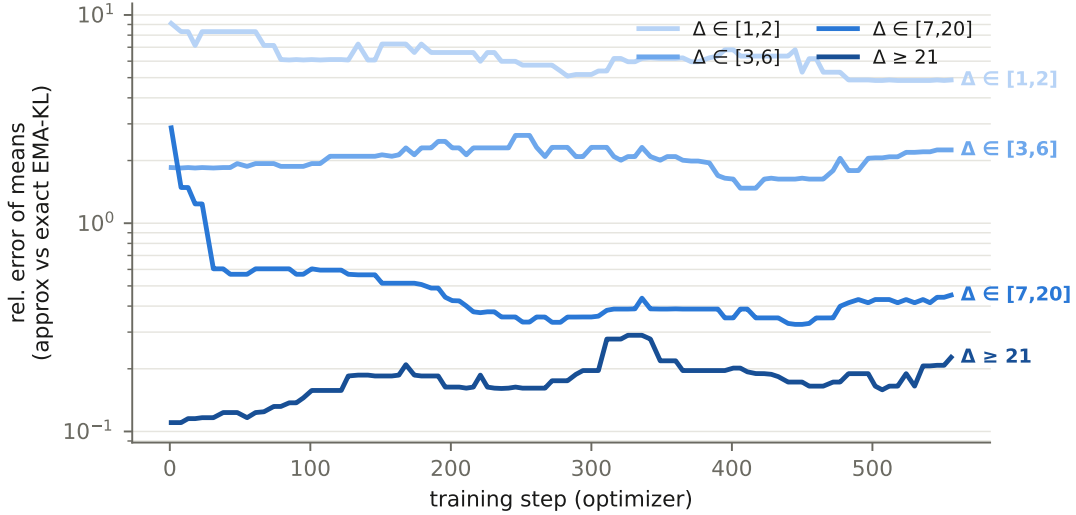


FIGURE 2. Relative error of the first-order surrogate against the exact EMA-KL (forwarded reference model), stratified by each token’s true lag Δ_i , over the decisive run (window-median smoothed; log scale). Sequential shading encodes the Δ ordering.

3. STATUS AND NEXT STEPS

Queued (cluster saturated ~ 40 h): uniform-dose sweep (`global_w0045/w015/w045`), completed $\beta=0$ heterogeneous and BF16 controls, FP4 $\beta=0$ ladder and rescue arms. Decisive comparisons to come: calibrated vs. *best* uniform dose (the dose-confound-proof test), and staleness- $\times$ -quantization interaction. Deferred by design (recorded in `EXPERIMENT_ROADMAP.md` §7): offset-rank rotation, common warm-start branching, ≥ 3 paired seeds on the two decisive arms.

Operational notes affecting reproduction: launchers now invoke `uv run --no-sync` (a node-local `uv` re-sync once recreated the shared venv mid-job and dropped `flash-attn`); and `recompute_logprobs` at `max_async_level=32` retains 33 CPU policy snapshots (~ 264 GB) and OOM-killed one control — all arms now share the vLLM anchor, which also removes an anchor-source confound.

APPENDIX A. CONFIGURATIONS

Common settings: `model=Qwen/Qwen3-4B-Instruct-2507`, `LR=1e-6`, `sampled_k=train_k=16`, `steps=720`, `type=tb`, `conf=COUNTDOWN.toml`, `seq_length=4096`, `eval_steps=20`, `reference_mode=ema`, `ema_alpha=0.9`, `kl_centering=false`, `vllm_logprobs=true`, batch size 512, wandb project `fp4_KL`, branch `fp4_KL`.

Arm (config file)	β	async	offsets	Δ -source
<code>homog10_approxKL_b005_vllmLP</code>	0.005	10	10,10,10,10	per_rollout
<code>hetero_approxKL_b005_perRollout</code>	0.005	32	1,4,10,32	per_rollout
<code>hetero_b0_async32</code>	0	32	1,4,10,32	—
<code>bf16_b0_async32</code>	0	32	—	—
<code>hetero_approxKL_global_w0045</code>	0.016	32	1,4,10,32	global
<code>hetero_approxKL_global_w015</code>	0.0533	32	1,4,10,32	global
<code>hetero_approxKL_global_w045</code>	0.16	32	1,4,10,32	global
<code>fp4sym_smoke</code> (60 steps)	0.005	10	—	global

KL penalty: approx path uses $c_i \text{clamp}(\log T_n - \log T_{n-\Delta_i}, \pm 10)$ summed over masked tokens, entering advantages as $-\beta(\cdot)$ (no centering). The uniform-dose arms hold one $w = \beta c_{\text{glob}}$ for every rollout with $c_{\text{glob}} = 0.9/(32 \cdot 0.1) = 0.28125$; w values bracket the calibrated arm’s mean dose over the four strata (≈ 0.0155). FP4 arms add `rollout_quant=nvfp4` (sampler QDQ at

checkpoint save) and `fake_quant=nvfp4` (trainer STE forwards; 252 patched linears = 36 layers $\times$ 7 projections; embeddings, `lm_head`, norms in BF16).

APPENDIX B. EVAL TABLE (SELECTED STEPS)

Step	100	200	300	400	500	600	680
Homog. $\Delta=10$ ref	0.731	0.754	0.773	0.799	0.806	0.823	0.828
Hetero + per-rollout c_i	0.725	0.764	0.811	0.799	0.815	—	—
Hetero $\beta=0$ (partial)	0.743	0.766	—	—	—	—	—

(Exact per-eval values in wandb; hetero arms end at steps 561 and 236 where their allocations/time limits ended.)